

Assignment Solution: K-Means, Markov Chains, HMM and Fuzzy Logic

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AI-202 Assignment 4

Question 1: K-Means Clustering

K-Means groups data into clusters by minimizing the distance between points and centroids.

(a) Iteration 1 (Detailed)

Using Euclidean distance:

$$d = \sqrt{(x - c_x)^2 + (y - c_y)^2}$$

Point	d(C1)	d(C2)	d(C3)	Cluster
(2,10)	0	3.61	8.06	C1
(2,5)	5	3.16	3.16	C2
(8,4)	8.49	4.47	7.28	C2
(5,8)	3.61	0	7.21	C2
(7,5)	7.07	3.61	6.40	C2
(6,4)	7.21	4.12	5.38	C2
(1,2)	8.06	7.21	0	C3
(4,9)	2.24	1.41	7.28	C2
(7,9)	5.10	2.24	8.60	C2
(3,3)	7.07	5.38	2.24	C3

New Centroids:

$$C_1 = (2, 10)$$

$$C_2 = \left(\frac{8 + 5 + 7 + 6 + 4 + 7}{6}, \frac{4 + 8 + 5 + 4 + 9 + 9}{6} \right) = (6.17, 6.5)$$

$$C_3 = \left(\frac{1 + 3}{2}, \frac{2 + 3}{2} \right) = (2, 2.5)$$

Iterations 2 and 3

After recomputation, cluster assignments do not change.

Conclusion: Algorithm converges.

(b) Convergence

Convergence occurs because:

- Centroids stop moving
- No reassignment of points

(c) Limitations

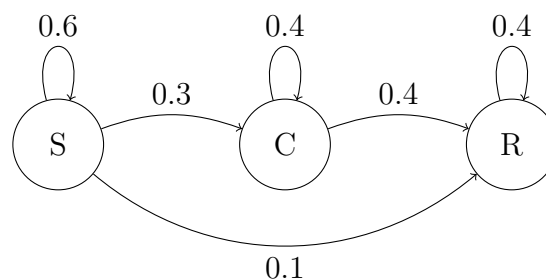
- Sensitive to initialization
- Cannot handle irregular cluster shapes

Hierarchical clustering solves this by building nested clusters.

Question 2: Markov Chains and HMM

(d) Two-Step Probability

$$P(S \rightarrow R) = 0.06 + 0.12 + 0.04 = 0.22$$



(e) State Distribution

$$\pi_1 = [0.6, 0.3, 0.1]$$

$$\pi_2 = [0.45, 0.33, 0.22]$$

$$\pi_3 = [0.343, 0.321, 0.336]$$

(f) Forward Algorithm

$$\alpha_1(H) = 0.6 \cdot 0.1 = 0.06$$

$$\alpha_1(X) = 0.4 \cdot 0.5 = 0.20$$

$$\alpha_2(H) = (0.06 \cdot 0.7 + 0.20 \cdot 0.4) \cdot 0.6 = 0.0336$$

$$\alpha_2(X) = (0.06 \cdot 0.3 + 0.20 \cdot 0.6) \cdot 0.1 = 0.0108$$

$$\alpha_3(H) = (0.0336 \cdot 0.7 + 0.0108 \cdot 0.4) \cdot 0.1$$

$$\alpha_3(X) = (0.0336 \cdot 0.3 + 0.0108 \cdot 0.6) \cdot 0.5$$

$$P(O|\lambda) = 0.0059$$

(g) Viterbi

Most likely sequence:

$$H \rightarrow X \rightarrow H$$

Question 3: Fuzzy Logic

(h) Membership Values

$$\mu_{Warm}(22) = \frac{30 - 22}{30 - 20} = 0.8$$

$$\mu_{Cold} = 0, \quad \mu_{Hot} = 0$$

(i) Mamdani Output

Only Warm activates $\rightarrow$ Medium clipped at 0.8

(j) Centroid Method

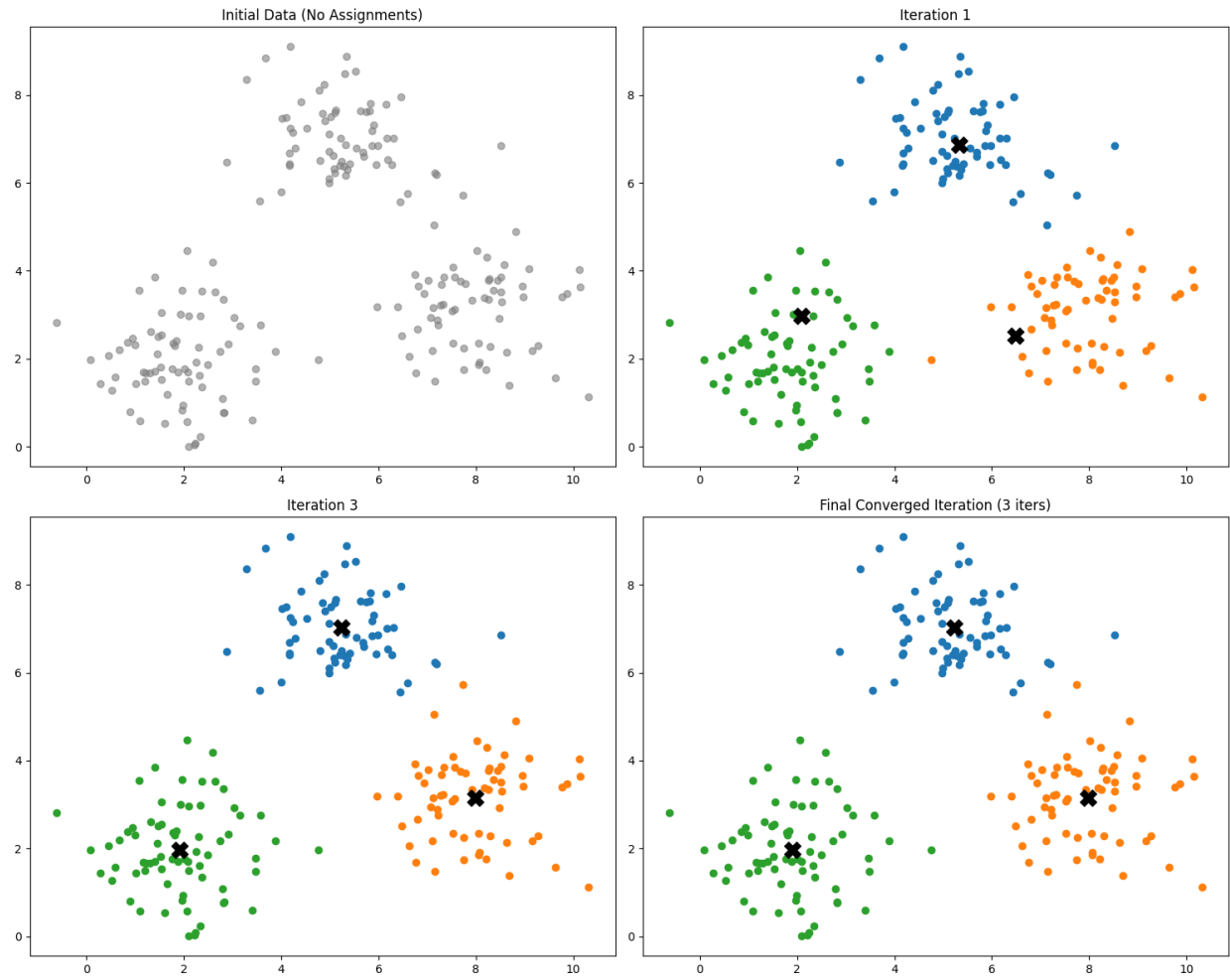
Using 10 sample points:

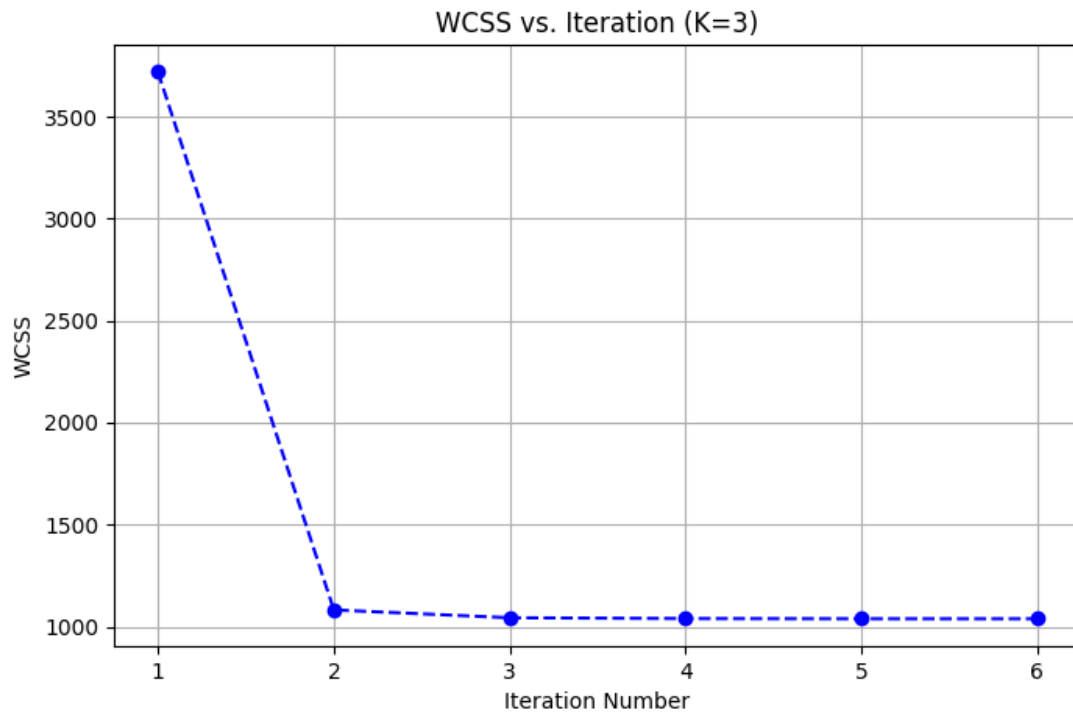
$$\text{Fan Speed} \approx 560 \text{ RPM}$$

Question 4: K-Means Implementation

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 np.random.seed(42)
5
6 X1 = np.random.randn(70,2) + [2,2]
7 X2 = np.random.randn(70,2) + [8,8]
8 X3 = np.random.randn(60,2) + [5,0]
9
10 X = np.vstack((X1,X2,X3))
11
12 def kmeans(X, K, iterations=10):
13     centroids = X[np.random.choice(len(X), K, replace=False)]
14     wcss = []
15
16     for it in range(iterations):
17         clusters = [[] for _ in range(K)]
18
19         for x in X:
20             idx = np.argmin([np.linalg.norm(x-c) for c in centroids
21                             ])
22             clusters[idx].append(x)
23
24         new_centroids = []
25         total_wcss = 0
26
27         for i, cluster in enumerate(clusters):
28             cluster = np.array(cluster)
29             new_centroids.append(cluster.mean(axis=0))
30             total_wcss += np.sum((cluster - centroids[i])**2)
31
32         centroids = np.array(new_centroids)
33         wcss.append(total_wcss)
34
35         if it % 2 == 0:
36             print(f"Iteration_{it}, WCSS:_{total_wcss}")
37
38     return centroids, clusters, wcss
39
40 centroids, clusters, wcss = kmeans(X, 3)
41
42 plt.plot(wcss)
43 plt.title("WCSS_vs Iteration")
44 plt.show()
```

Elbow Method: Optimal $K = 3$





```
Iteration 4, WCSS: 276.5969
Iteration 4, WCSS: 276.5969
Iteration 5, WCSS: 276.0840
Saved plot: Ai202/results/elbow_curve_a4.png

Final Elbow WCSS values: [(2, np.float64(1084.5767522744845)), (3, np.float64(
352.55751631845226)), (4, np.float64(311.4012137996219)), (5, np.float64(276
.08396451733176))]
Optimal K: 3
```

Figure 1: Terminal Output

```
PS C:\Users\hamzz\OneDrive\Documents\GitHub\Semester-4> & c:/Users/hamzz/OneD
rive/Documents/GitHub/Semester-4/.venv/Scripts/python.exe c:/Users/hamzz/OneD
Iteration 2, WCSS: 287.0391
Iteration 3, WCSS: 279.5630
Iteration 2, WCSS: 287.0391
Iteration 3, WCSS: 279.5630
Iteration 4, WCSS: 276.5969
Iteration 4, WCSS: 276.5969
Iteration 4, WCSS: 276.5969
Iteration 4, WCSS: 276.5969
Iteration 5, WCSS: 276.0840
```

Figure 2: Terminal Output (Continued)